

I. Create the following folders:

PRISMA FLOW DIAGRAM (add PRISMA diagram, download it from Google, word, the first option)

CONFERENCE PAPER (add IEEE manuscript, download it from Google, word)

INCLUDED (you will add a minimum of 22 articles that will be “included”)

TABLES (Have word documents: MAIN TABLE, DERIVED TABLE, SEPARATE TABLES )

GRAPHS

SEARCH STRATEGY (add word document with search strategy, search strategy. This is where you will save keywords, database, link, results, search query and research questions)

Your IEEE manuscript template must be customized to have the following skeleton:

**Title, authors .....as shown in the template**

**Abstract**

**Introduction**

**Research Methodology**

**Literature Review**

**Synthesis and analysis**

**Results and discussion**

**Challenges and opportunities**

**SWOT**

**Summary of articles**

**Conclusion**

**Acknowledgement**

**Appendix**

2. Search the articles
3. Screen by abstract (you are expected to be left with around 22-40 articles)
4. Screen by retrieval, retrieve those 22-40 articles and save them under the “included” folder
5. Screen by reviewing, construct research questions,
6. Draft research questions:

**RQ1:** What are the commonly used features in irrigation methods and what is the data sources type used?

**RQ2:** "What are the methods used to develop informative communication platforms for farmers in rural areas, particularly considering challenges like limited connectivity?

**RQ3:** What is the most effective performance metrics used for evaluating an AI/ML chatbot in smart agriculture, particularly considering their application in crop management and recommendation systems?

**RQ4:** What are the limitations of an AI model concerning data collection, connectivity?

7. Use ChatPDF to the articles the following table gives an overview of how you can review articles in SLR:

**MAIN TABLE**

| Authors                                                                                     | Problem identified                                                                                                                           | Dataset                                                  | Method                                                                                                                                                                           | Performance metrics                                                                                                                                                                         | Findings                                                                                                                                                                         | Limitations                                                                                                  |
|---------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Stefano Leonori, Stefano Mattei, Luigi Anniballi, and Fabio Massimo Frattale Mascioli, 2024 | The need for autonomous, energy-efficient, and minimally invasive farming systems to enhance productivity and sustainability in agriculture. | Statistics and Research areas                            | Hierarchical control system with ROS and FSM, integrating GNSS-based navigation with RTK correction, establishing robust communication networks (Ethernet, CAN, I2C, USB, UMTS), | Accuracy = <20 cm + RTK GNSS<br><br>obstacle detection within 4 m using sonar,<br><br>energy consumption per cycle =53.1 Wh<br><br>total cycle time =462 sec<br><br>Max H-Cart speed=1 km/h | Electric multi-robot agribot system enhanced precision farming by reducing soil compaction and energy consumption. Improved operational safety through cable-way transportation. | Relying on the current prototype systems which is far from achieving full autonomy and commercial viability. |
| Sehrish Munawar Cheema and                                                                  | The limited accuracy of soil fertility recommendations due to insufficient                                                                   | Kaggle, specifically the "Soil fertility dataset," which | Real-time soil and environmental data collection                                                                                                                                 | Accuracy= 98%<br>Precision =0.99<br>Recall = 0.98)<br>F1-score =0.98                                                                                                                        | The system achieved high accuracy up to 98% and                                                                                                                                  | Varied soil properties across regions affecting                                                              |

|                                                                                                                                          |                                                                                                           |                                                                                                                                      |                                                                                                                                                                 |                                                                                                                                                           |                                                                                                                                                       |                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| Ivan Miguel Pires .2024                                                                                                                  | sensor data and regional variability                                                                      | contains 2200 instances and 7 data fields: N, P, K, pH, T (Temperature), H (Humidity), and Rainfall                                  | via sensors, transmission to cloud, + machine learning models for crop recommendation                                                                           |                                                                                                                                                           | balanced precision, recall, and F1-score for crop prediction                                                                                          | accuracy, incomplete implementation of some models, manual data entry, and missed parameters during training                                       |
| Lei Liu, Fan Yang, Xiangyi Liu, Yuefeng Du (corresponding author), Xiaoyu Li, Guorun Li, Du Chen, Zhongxiang Zhu, and Zhenghe Song, 2024 | Low sensing accuracy, poor operability, and low operating efficiency of agricultural field robots         | Robots + Sensors + Technologies                                                                                                      | Bibliometric analysis and comparative review of technological features + applications of agricultural robots.                                                   | sensing accuracy, operability, operating efficiency, high-throughput data collection, and autonomous decision-making capabilities                         | Recent advances in agricultural robots include drone-based yield estimation, integrated sensing, autonomous navigation, and targeted crop management. | Agricultural drones limited range, adaptive navigation challenges, payload capacity, propeller noise, and inability to fully replace ground robots |
| Aglaia Liopa-Tsakalidi, Vasileios Thomopoulos, Pantelis Barouchas, Achilles D. Boursianis, and Sotirios K. Goudos . 2024                 | Inefficient water use in traditional irrigation practices leading to waste and sustainability challenges. | Soil moisture, Temperature, Irrigation volume, battery voltage, and connectivity metrics collected over 2019–2021 in an olive grove. | LoRaWAN + IoT sensor nodes, and automate water management in the olive grove, with data transmitted to a cloud-based platform for analysis and decision-making. | Range up to 10–40 km in rural zones<br>Battery life of several months= 42% reduction in water consumption in 2020, 25% reduction in 2021 compared to 2019 | The system achieved Significant water savings =42% in 2020 and 25% in 2021 and enabled real-time, cost-effective smart irrigation for olive groves    | LoRaWAN range and Antenna size limitations affect communication quality in obstacle-rich environments.                                             |

8. Derive the second table from the first table, the second table will be as follows:

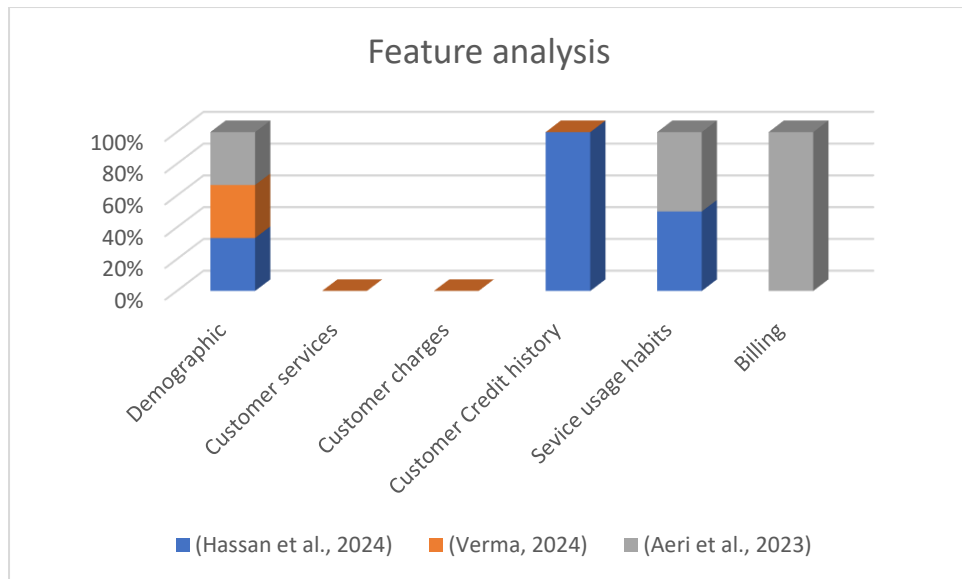
| Authors               | Feature selection     | Dataset   |          | Method     |              |               | Performance metrics |               |               |              | Limitations          |
|-----------------------|-----------------------|-----------|----------|------------|--------------|---------------|---------------------|---------------|---------------|--------------|----------------------|
|                       |                       | Real-Time | Historic | Supervised | Unsupervised | Deep Learning | Acc %               | Rec           | Prec          | F1-score     |                      |
| Mazibuko et al., 2024 | Soil type, Plant type | x         | √        | x          | √            | x             | LR=78               | KNN = 98      | SVM =76       | N/A          | Network Constraints  |
| Nhleko, 2025          |                       | √         | x        | √          | x            | √             | FB= 88              | ANN = 97      | LSTM=67       | SVM=88       | Limited to English   |
|                       |                       |           |          |            |              |               |                     |               |               |              |                      |
| Results               | Common = soil type    | 1/2= 50%  | 1/2 =50% | 1/2 =50%   | 1/2 =50%     | 0/2=0%        | Max= FB=88 %        | Max= KNN=98 % | Max= SVM=76 % | Max=88 % SVM | Common= Data quality |

9. Separate tables and draw graphs

Add all authors and features in Excel and draw a graph to get the commonly used feature. Also, for all authors and limitations

| Author                | Soil type | Plant type | Sensor humidity | Temperature | Emotions | Warnings |
|-----------------------|-----------|------------|-----------------|-------------|----------|----------|
| Mazibuko et al., 2024 | 1         | 0          | 0               | 1           | 1        | 0        |
| Nhleko, 2025          | 1         | 0          | 0               | 0           | 0        | 0        |
| (Aeri et al., 2023)   | 1         |            |                 |             | 1        | 1        |
|                       |           |            |                 |             |          |          |

Then draw Bar graphs or any suitable graphs, e.g



10. Sign up for NotebookLM (AI Tools), create a notebook, and upload ALL articles under “INCLUDED”

11. Now it's time to write..... You write a command for each section. E.g.

“Write a concise three-paragraph introduction using the selected articles. The introduction should include the context of the study, the research problem, and the research questions, as well as justify why a systematic literature review is used and what needs to be done to address the research problem. Please use the Harvard referencing style.”

Then move to research methodology

“Write a concise research methodology, using the selected PRISMA flow, and include the following subsections: search strategy, inclusion and exclusion criteria, synthesis frame, and data extraction. “

Then move to your literature review. Discuss each table and graph with the help of NotebookLM. And so on....

12. PRISMA Flow Diagram

